I Introduction

A) Assume wire is ideal conduction ∴ the potential difference (V) accross the resistor equals the applied voltage of the battery

II Series Resistors

$$R_T = R_1 + R_2 + \cdots + R_N \quad (1)$$

III Series Circuit

$$I_s = \frac{E}{R_T} \quad (3)$$

IV Power Distribution in Series Circuit

$$P_E = P_{R_1} + P_{R_2} + \cdots + P_{R_N} \quad (7)$$

Where $P_{R_N} = V_N I_N = I_N^2 R_N = \frac{V_N^2}{R_N}$

V Voltage Sources In Series

A) If the are in "the same direction" then

$$E_T = E_1 + E_2 + \cdots + E_N$$

VI Kirchoff's Voltage Law

The Algebraic sum of the potentials rises and drops around a closed paths is zero

$$\sum_{\circlearrowleft} V = 0 \quad (8)$$

The applied voltage of a series of dc circuits will equal the sum of the drop voltage of the circuit

$$\sum_{\circlearrowleft} V_{rises} = \sum_{\circlearrowleft} V_{drops}$$

VII Voltage Devision In A Series

$$V_x = R_x \frac{E}{R_T}$$

(10)